

## Research Areas

Vector Databases, Data Agents, Data Systems for AI Agents

## Employment

2024-present      Assistant Professor - The University of Texas at Austin  
2021-2023        Postdoctoral Scholar - University of California, Berkeley  
                      Advisor: Aditya G. Parameswaran, Associate Professor

## Education

2015-2020        Ph.D. in Computer Science - University of Chicago  
                      Advisor: Aaron J. Elmore, Associate Professor  
2011-2014        M.S. in Computer Science - Institute of Computing Technology, Chinese Academy of Sciences  
                      Advisor: Wei Li, Associate Professor  
2007-2011        B.S. in Software Engineering - Huazhong University of Science & Technology

## Publications

**P22.** Optimal Predicate Pushdown Synthesis

Robert Zhang, Eric Hayden Campbell, **Dixin Tang**, Isil Dillig  
**PLDI 2026**

**P21.** Homomorphism Calculus for User-Defined Aggregations

Ziteng Wang, Ruijie Fang, Linus Zheng, **Dixin Tang**, Isil Dillig  
**OOPSLA 2025**

**P20.** Transactional Panorama: A Conceptual Framework for User Perception in Analytical Visual Interfaces (extended version)

**Dixin Tang**, Alan Fekete, Indranil Gupta, Aditya G. Parameswaran  
**VLDBJ 2025** (The "Best of VLDB 2023" special issue of VLDB Journal)

**P19.** Tigon: A Distributed Database for a CXL Pod

Yibo Huang, Haowei Chen, Newton Ni, Yan Sun, Vijay Chidambaram, **Dixin Tang**, Emmett Witchel  
**OSDI 2025**

**P18.** Impeller: Stream Processing on Shared Logs

Zhiting Zhu, Zhipeng Jia, Newton Ni, **Dixin Tang**, Emmett Witchel  
**EuroSys 2025**

**P17.** Pasha: An Efficient, Scalable Database Architecture for CXL Pods

Yibo Huang, Newton Ni, Vijay Chidambaram, Emmett Witchel, **Dixin Tang**  
**CIDR 2025**

**P16.** Dealing with Acronyms, Abbreviations, and Typos in Real-World Entity Matching

Joshua Wu, **Dixin Tang**, Nithin Chalapathi, Tristan Chambers, Julie Ciccolini, Cheryl Phillips, Lisa Pickoff-White, Aditya G. Parameswaran  
**VLDB 2024, Industrial Track**

**P15.** Visualizing Spreadsheet Formula Graphs Compactly

Fanchao Chen, **Dixin Tang**, Haotian Li, Aditya G. Parameswaran  
**VLDB 2023, Demo**

- P14. Transactional Panorama: A Conceptual Framework for User Perception in Analytical Visual Interfaces**  
**Dixin Tang**, Alan Fekete, Indranil Gupta, Aditya G. Parameswaran  
**VLDB 2023 ("Best of VLDB 2023")**
- P13. Efficient and Compact Spreadsheet Formula Graphs**  
**Dixin Tang**, Fanchao Chen, Christopher De Leon, Tana Wattanawaroon, Jeaseok Yun, Srinivasan Seshadri, Aditya G. Parameswaran  
**ICDE 2023**
- P12. Flexible Rule-Based Decomposition and Metadata Independence in Modin: A Parallel Dataframe System**  
Devin Petersohn\*, **Dixin Tang\***, Rehan Durrani, Areg Melik-Adamyan, Joseph E. Gonzalez, Anthony D. Joseph, Aditya G. Parameswaran  
**VLDB 2022 (\*Equal contribution)**
- P11. Lux: Always-on Visualization Recommendations for Exploratory Dataframe Workflows**  
Doris Jung-Lin Lee, **Dixin Tang**, Kunal Agarwal, Thyne Boonmark, Caitlyn Chen, Jake Kang, Ujjaini Mukhopadhyay, Jerry Song, Micah Yong, Marti A. Hearst, Aditya G. Parameswaran  
**VLDB 2022**
- P10. Enhancing the Interactivity of Dataframe Queries by Leveraging Think Time**  
Doris Xin, Devin Petersohn, **Dixin Tang**, Yifan Wu, Joseph E. Gonzalez, Joseph M. Hellerstein, Anthony D. Joseph, Aditya G. Parameswaran  
**IEEE Data Eng. Bull. 2021**
- P9. Resource-Efficient Shared Query Execution via Exploiting Time Slackness**  
**Dixin Tang**, Zechao Shang, William Ma, Aaron J. Elmore, Sanjay Krishnan  
**SIGMOD 2021**
- P8. CIAO: An Optimization Framework for Client-Assisted Data Loading**  
Cong Ding, **Dixin Tang**, Xi Liang, Aaron J. Elmore, Sanjay Krishnan  
**ICDE 2021, Short Paper**
- P7. CrocodileDB in Action: Resource-Efficient Query Execution by Exploiting Time Slackness**  
**Dixin Tang**, Zechao Shang, Aaron J. Elmore, Sanjay Krishnan, Michael J. Franklin  
**VLDB 2020, Demo**
- P6. Thrifty Query Execution via Incrementability**  
**Dixin Tang**, Zechao Shang, Aaron J. Elmore, Sanjay Krishnan, Michael J. Franklin  
**SIGMOD 2020**
- P5. CrocodileDB: Efficient Database Execution through Intelligent Deferment**  
Zechao Shang, Xi Liang, **Dixin Tang**, Cong Ding, Aaron J. Elmore, Sanjay Krishnan, Michael J. Franklin  
**CIDR 2020**
- P4. Intermittent Query Processing**  
**Dixin Tang**, Zechao Shang, Aaron J. Elmore, Sanjay Krishnan, Michael J. Franklin  
**VLDB 2019**
- P3. Socrates: The New SQL Server in the Cloud**  
Panagiotis Antonopoulos, Alex Budovski, Cristian Diaconu, Alejandro Hernandez Saenz, Jack Hu, Hanuma Kodavalla, Donald Kossmann, Umar Farooq Minhas, Naveen Prakash, Hugh Qu, Chaitanya Sreenivas Ravella, Krystyna Reisteter, Sheetal Shrotri, **Dixin Tang**, Vikram Wakade  
**SIGMOD 2019**
- P2. Toward Coordination-Free and Reconfigurable Mixed Concurrency Control**  
**Dixin Tang**, Aaron J. Elmore  
**USENIX'ATC 2018**
- P1. Adaptive Concurrency Control: Despite the Looking Glass, One Concurrency Control Does Not Fit All**  
**Dixin Tang**, Hao Jiang, Aaron J. Elmore  
**CIDR 2017**

## Team

### PhD Students

- Abhinav Gyawali (co-advised with Vijay Chidambaram)
- Jiaqi Huang
- Hexu Li
- Shurang Wu (co-advised with Vijay Chidambaram)
- Tianming Wu
- Yibo Huang (co-advised with Prof. Witchel)
- Ziteng Wang (co-advised with Prof. Dillig)

### Undergraduate and Master's Students

- Yingfei Tang

## Honors and Awards

- Distinguished Reviewer Award, SIGMOD 2025
- Best of VLDB 2023

## Professional Services

### Program Committee

- SIGMOD - 2022, 2023, 2025, 2026, 2027
- VLDB - 2025, 2026, 2027
- ICDE - 2025, 2026
- EDBT - 2024
- SIGMOD Demo Track - 2022, 2024, 2025
- HILDA - 2023, 2024

### Conference Reviewer

- IEEE VIS - 2021

### Journal Reviewer

- VLDB Journal
- Distributed and Parallel Databases Journal

## Teaching

### The University of Texas at Austin

- Instructor - CS 395T: Advanced Database Systems, Fall 2026
- Instructor - CS 378: Database System Design, Spring 2026
- Instructor - CS 395T: Database Systems and LLMs, Fall 2025
- Instructor - CS 347: Data Management, Fall 2024
- Instructor - CS 395T: Advanced Query Optimizations, Spring 2024

### University of Chicago

- TA - CMSC 23500 - Introduction to Database Systems, Winter 2017-2020
- TA - MPCS 52040 - Distributed Systems, Spring 2016
- TA - MPCS 51040 - C Programming, Fall 2015

## Industry Experience

### **Internship at Microsoft Research**

June 2018-Sep. 2018

*Project:* Benchmarking Socrates

*Mentor:* Umar Farooq Minhas

Socrates is a new cloud-native database that decouples computation from storage. My internship involved testing the new database architecture of Socrates in an industrial setting, understanding its performance bottlenecks, and proposing optimization opportunities.